

Corporate Network Infrastructure Proposal

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Networking & Cybersecurity AAS Independent Capstone Project

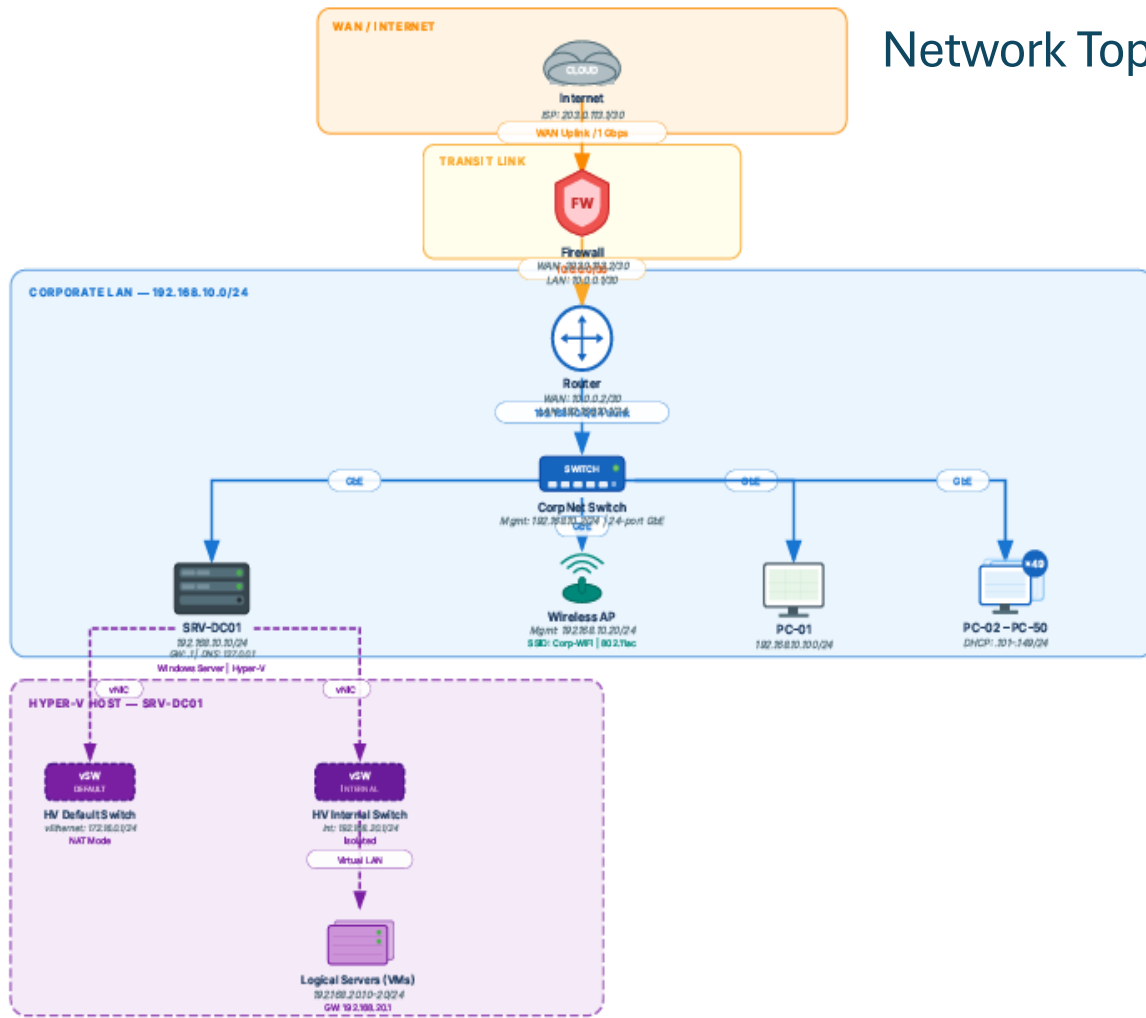
Scenario

For this project, I have designed and built an enterprise network for a medium-sized law firm that is transitioning to a fully digital office and discontinuing the use of paper for client files.

Project Scope

Design and build a corporate network infrastructure for 50 users within a nested Hyper-V environment, implement Windows Server corporate domain, network infrastructure, establish network share servers, create full subnet plan, deploy security plans for incident response and backup/recovery, document and present the project budget and timeline for completion.

Network Topology



Rev 1.0 - Aug 2026

DEVICE TYPES



IP ADDRESS SCOPE

ZONE	NETWORK	SUBNET MASK	RANGE	GATEWAY
WAN	203.0.113.0	/30 (255.255.255.252)	.1-.2	ISP
Transit	10.0.0.0	/30 (255.255.255.252)	.1-.2	—
Corporate LAN	192.168.10.0	/24 (255.255.255.0)	.1-.254	192.168.10.1
DHCP Pool	192.168.10.0	/24	.101-.149	192.168.10.1
HV Default (NAT)	172.16.0.0	/24 (255.255.255.0)	.1-.254	172.16.0.1
HV Internal	192.168.20.0	/24 (255.255.255.0)	.10-.20	192.168.20.1

DHCP Scope

- Network: 192.168.10.0/24
- Scope Range: 192.168.10.50 – 192.168.10.200
- Exclusion Range: 192.168.10.1 – 192.168.10.49
- Purpose: Addresses for PC 01 through PC 50 and other dynamic LAN clients

Remote Access

RRAS PPTP VPN has been configured for remote access to the domain servers for authorized users. VPN users receive IP addresses from the RRAS pool, not DHCP scope.

The Firewall VPN rules allow for PPTP and blocks direct external access to any network devices. Outbound traffic to the web is filtered through outbound rules.

Only RBAC group, “Remote Access Users” can use VPN, and domain credentials are required to login to the VPN.

Security Plan

Perimeter firewall is installed for controlling inbound and outbound network traffic. Further, RBAC has been implemented via Active Directory. The core EDR is lightweight and functions for malware prevention, threat detection, and event logging.

Backup & Recovery

To ensure business continuity and the recoverability of systems, configurations and data, the domain network’s critical services must be protected from loss. These services include AD DS, DNS, DHCP, RRAS, and our File/Print/SQL servers.

Frequency: Daily incremental backup, weekly full backup, monthly archive backup

Recovery procedure:

Servers: restore system, reapply all service configs, recover shared service roles, then confirm domain is functional.

Clients: re-flash workstations, rejoin domain, restore user profile and data

Data Recovery: restore file shares, SQL databases and logs, validate access controls and integrity of data

Project Budget

Item	Role / Description	Category	Low Estimate	High Estimate
HARDWARE				
Cisco Firepower 1010	Edge firewall / NGFW	Hardware – Security	\$800	\$1,500
Cisco Catalyst 9300	Core/distribution switch	Hardware – Switching	\$5,000	\$10,000
Cisco Catalyst 9200	Access layer switch	Hardware – Switching	\$2,500	\$5,000
Cisco 150AX Access Point	Wi-Fi 6 wireless access point	Hardware – Wireless	\$400	\$800
Cisco UCS C220 Server	Rack-mount server / compute	Hardware – Compute	\$8,000	\$20,000
Hardware Subtotal			\$16,700	\$37,300
LICENSING				
Cisco Duo MFA	Multi-factor authentication (100 users/yr)	Licensing – Security	\$3,600	\$10,800
Cisco Secure Endpoint	Endpoint detection & response (100 endpoints/yr)	Licensing – Security	\$5,000	\$12,000
Cisco Firepower Licensing	IPS / URL filtering / AMP subscription	Licensing – Security	\$1,500	\$4,000

Item	Role / Description	Category	Low Estimate	High Estimate
Licensing Subtotal			\$10,100	\$26,800
GRAND TOTAL — Year 1			\$26,800	\$64,100

Implementation Timeline/Schedule

Work Produced	Week	Tasks
Planning	Week 1	Requirements Gathering; Network Design; IP Addressing Plan; Budget & Device Selection
Virtualization & Server Deployment	Week 2	Hyper-V Setup; SRV-DC01 Deployment; AD DS/DNS/DHCP Setup; RRAS Installation
Infrastructure Setup	Week 3	Firewall Configuration; Router & Switch Setup; Wireless AP Integration; Subnet Plan Validation
Security Configuration	Week 4	EDR Deployment; Firewall Policy Tuning; RBAC & MFA Integration; Incident Response Documentation
Client Deployment & Testing	Week 5	PC-01 to PC-50 Domain Join; GPO Validation; Connectivity Testing
Final Documentation	Week 6	Network Diagram; Security Documentation; Budget & Schedule; Backup & Recovery Plan; Final Presentation